Google Developer Group
Bandung Institute of Technology

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| **Team name** | Tide Ledger |
| **Project title** | RapiLPJ: Pre-Submission Audit for Student Organization Activity Reports |
| **Selected case** | Case 1: Smarter Campus and Learning |
| **Members** | Vincentius Bryan Kwandou (leader), Valentino Filemon Limer, Putra Purwanugraha Tengbunan, Marvel Harjosetio |


## Abstract

**1. Problem.** Student treasurers must complete an accountability report (LPJ) before faculty release subsequent event funding. Rules differ by campus: original ink stamps, meal attendance sheets, matching ledger totals, revenue stamps above Rp5,000,000 (Law 10/2020), tax withholding, and strict deadlines. Treasurers lack formal accounting training, causing recurring rework. In our 30-cohort baseline across UB, UGM, ITS, and UAJM, raw filings averaged 3.7 mechanical defects, causing 100% of sampled submissions to face initial desk-review rejection or mandatory revisions and 7 to 14 days of funding delay.

**2. Intended Users.** Primary users are student treasurers and secretaries of department associations, BEM, and clubs. Faculty reviewers conduct desk-checks. RapiLPJ does not disburse funds.

**3. Supporting Observations or Evidence.** FIB UB requires stamped originals, meal attendance sheets, and matching totals. FEB UGM enforces a Rp10,000 stamp above Rp5,000,000 and digital scans for thermal paper. ITS departments set 7 to 10 day deadlines and itemized receipt rows. A 2021 UAJM study documents unstandardized activity reporting. FILKOM UB 2026 circulars report recurring tax withholding and completeness failures.

**4. Proposed Solution and Core User Flow.** RapiLPJ acts as a digital practice run: rejection happens on a laptop before reaching the faculty desk. The user flow follows five steps: (1) receipt upload, (2) Gemini 2.0 Flash perceptual extraction into structured JSON with a confidence score (below 0.85 alerts treasurers and triggers mandatory scan uploads for thermal slips), (3) side-by-side treasurer confirmation where every manual change enters a dual-record audit trail exposing overrides to reviewers, (4) client-side kernel deterministic audit checking duplicates, event timeline sanity, statutory stamps, and budget variance, and (5) Gemini Advisory translating findings into plain-language guidance and official addendum drafts. Rule governance is managed by a versioned JSON card authored by student affairs and endorsed by two faculty advisors via an anchor hash that prevents unauthorized rule tampering. The validated dossier is locked with a SHA-256 seal, enabling instant zero-server offline desk-checks by faculty. Offline manual data entry serves as a zero-network fallback.

**5. How It Differs from Existing Alternatives.** A raw LLM miscalculates arithmetic. Spreadsheets permit invalid stamps, guessed taxes, and silent edits without detection. Campus portals collect files only after submission, discovering defects too late to prevent frozen funding.

**6. Planned Technologies.** Google AI Studio and Vertex AI (Gemini 2.0 Flash Multimodal Vision and Governance Advisory), Firebase Hosting, Google Sheets compatible CSV export, client JavaScript kernel, FileReader, and Web Crypto SHA-256. Files and JSON remain strictly in client volatile memory and local .lpj files with zero cloud database storage, ensuring complete student fiscal data privacy. WCAG 2.1 AA.

**7. Expected Benefit and Development Plan.** In a 30-cohort synthetic simulation benchmark, the kernel resolved all 112 mechanical defects (13 missing stamps, 15 thermal slips, 20 duplicate invoices) in 15.6 ms per dossier. A one-semester pilot across 20 active student organizations targets reducing desk-check revision cycles from a baseline of 3.4 to at most 1 cycle, eliminating funding freezes. Arithmetic never trusts the model.



**Source Acknowledgments.** [1] P. Siada & A. L. Tungadi, TEMATIKA J. Penelit. Tek. Inform. Sist. Inf. UAJM, 2021, doi:10.56963/tematika.v9i1.119 (studi sistem informasi portofolio kemahasiswaan BKAM UAJM). [2] Pedoman LPJ, FIB Universitas Brawijaya, fib.ub.ac.id. [3] Prosedur Proposal & LPJ SIPRO, FEB UGM & UU Bea Meterai No. 10/2020. [4] SOP HIMA, Desain Interior ITS, its.ac.id. [5] SOP LPJ Mahasiswa Biologi, ITS, its.ac.id. [6] Surat Edaran Mekanisme LPJ Ormawa, FILKOM UB, 2026.